*SaltProc verification with OpenMC

*REPORT PARTICIPANTS:

Oleksandr R Yardas [1], Guillaume Giudicelli[3], Madicken Munk [1]
1 University of Illinois Urbana-Champaign
2 Idaho National Lab

*SCIENTIFIC ACHIEVEMENT:

Accurately simulating fuel reprocessing in molten salt reactors is critical to understanding long term behavior of the flux and material compositions. SaltProc is a tool that allows the simulation of batchwise resprocessing of fuel salt in molten salt reactors. In this project, I used OpenMC to perform depletion calculations, which then coupled to SaltProc to account for online reprocessing. These results were compared with Serpent2 depletion simulations coupled to SaltProc on the same model. I found good agreement between the two codes in both k-effective and isotopic composition of the fuel.

*SIGNIFICANCE:

Molten salt reactors (MSRs) face several technical and logistical challenges before they can be licensed, constructed, and ultimately deployed for commercial power production. Modeling and simulation will play a critical role in licensing MSRs and other advanced reactors. In preparation for MSR deployment, both the DOE-NE and the NRC identified several technical gaps in current modeling and simulation tools that are necessary for license application reviews. In particular, accurate and precise calculation of the fuel salt composition over time in MSRs is paramount to the continued safe operation and decommissioning of these reactors. The validation of SaltProc with OpenMC gives researchers an additional pathway in SaltProc a to study the fuel composition and evolution in MSRs.

*KEY PUBLICATIONS:

Yardas, O. R. (2023). *Implementation and validation of OpenMC depletion capabilities in SaltProc* [Thesis, University of Illinois at Urbana-Champaign]

*SPONSOR/PROGRAM:

US NRC Integrated University Grant Program Fellowship
Cody Permann

***IMAGES AND CAPTION:**

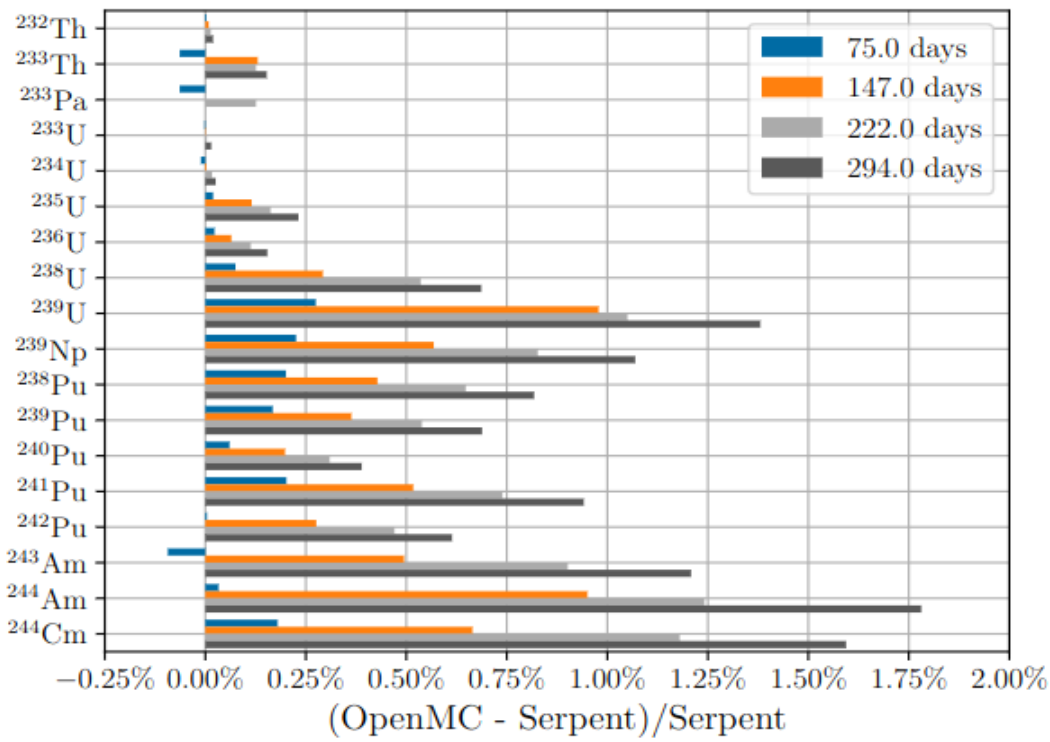

Fig 1: Relative difference of actinides mass between OpenMC and Serpent2 coupled with SaltProc.

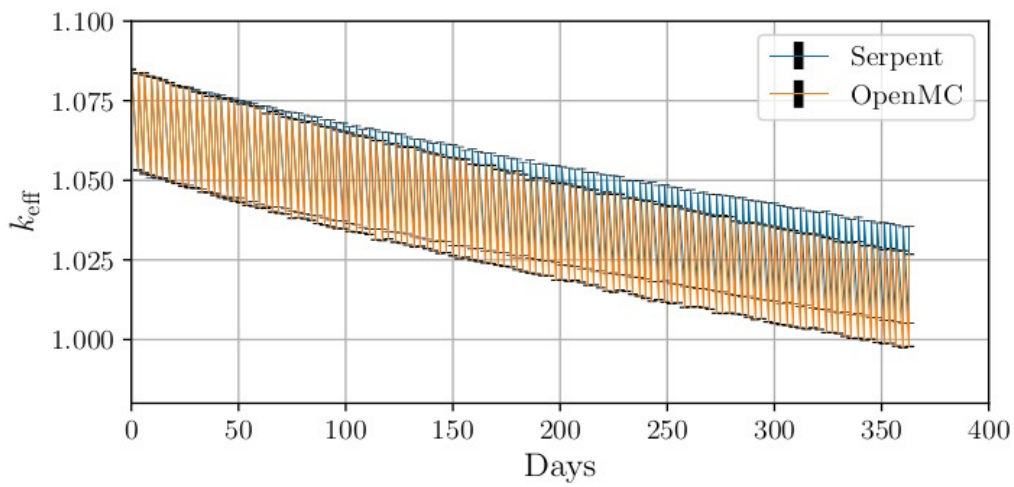

Fig 2: Comparison of K-effective between OpenMC and Serpent2 coupled with SaltProc.